In the 4h version.

4.38 With reference to the 2-D discrete convolution theorem in Eqs. (4-95) and (4-96) (entry 6 in Table 4.4), show that

(a) $(f \star h)(x,y) \Leftrightarrow (F \bullet H)(u,v)$

(b)* $(f \bullet h)(x,y) \Leftrightarrow (1/MN)[(F \star H)(u,v)]$

(*Hint:* Study the solution to Problem 4.18.)

4.48* A continuous Gaussian lowpass filter in the continuous frequency domain has the transfer function

$$H(\mu,\nu) = Ae^{-(\mu^2+\nu^2)/2\sigma^2}$$

Show that the corresponding filter kernel in the continuous spatial domain is

$$h(t,z) = A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)}$$

4.49 Given an image of size $M \times N$, you are asked to perform an experiment that consists of repeatedly lowpass filtering the image in the frequency domain using a Gaussian lowpass filter transfer function with a cutoff frequency, D_0. You may ignore computational round-off errors.

(a)* Let K denote the number of applications of the filter. Can you predict (without doing the experiment) what the result (image) will be for a sufficiently large value of K? If so, what is that result?

4.56* Show how the Butterworth highpass filter transfer function in Eq. (4-121) follows from its lowpass counterpart in Eq. (4-117).

数字图像处理 作业4

南大 自动化学院
202331221b

4.38 With reference to the 2-D discrete convolution theorem in Eqs. (4-95) and (4-96) (entry 6 in Table 4.4), show that

(a) $(f \star h)(x,y) \Leftrightarrow (F \cdot H)(u,v)$

(b)* $(f \cdot h)(x,y) \Leftrightarrow (1/MN)[(F \star H)(u,v)]$

(*Hint:* Study the solution to Problem 4.18.)

DFT:

$$F(u,v)=\mathcal{F}[f(x,y)]=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} f(x,y)\, e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$f(x,y)=\mathcal{F}^{-1}[F(u,v)]=\frac{1}{MN}\sum_{u=0}^{M-1}\sum_{v=0}^{N-1} F(u,v)\, e^{j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

(a) $$f(x,y)*h(x,y)=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)\, h(x-m,y-n)$$

Let $c(x,y)=f(x,y)*h(x,y)$

$$C(u,v)=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} c(x,y)\, e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$=\sum_{x=0}^{M-1}\sum_{y=0}^{N-1}\left[\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)\, h(x-m,y-n)\right] e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1}\left[f(m,n)\sum_{x=0}^{M-1}\sum_{y=0}^{N-1} h(x-m,y-n)\, e^{-j2\pi\left(\frac{u(x-m)}{M}+\frac{v(y-n)}{N}\right)} \cdot e^{-j2\pi\left(\frac{um}{M}+\frac{vn}{N}\right)}\right]$$

$$=\sum_{m=0}^{M-1}\sum_{n=0}^{N-1} f(m,n)\cdot e^{-j2\pi\left(\frac{um}{M}+\frac{vn}{N}\right)} \cdot H(u,v) = F(u,v)\cdot H(u,v)$$

(b) Let $p(x,y)=f(x,y)\cdot h(x,y)$ and compute its DFT:

$$P(u,v)=\sum_{x,y} f(x,y)\, h(x,y)\, e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$=\sum_{x,y}\left[\frac{1}{MN}\sum_{\alpha,\beta} F(\alpha,\beta)\, e^{j2\pi\left(\frac{\alpha x}{M}+\frac{\beta y}{N}\right)}\right]\left[\frac{1}{MN}\sum_{\gamma,\delta} H(\gamma,\delta)\, e^{j2\pi\left(\frac{\gamma x}{M}+\frac{\delta y}{N}\right)}\right] e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)}$$

$$=\frac{1}{MN}\sum_{\alpha,\beta}\sum_{\gamma,\delta} F(\alpha,\beta)\, H(\gamma,\delta)\underbrace{\sum_{x} e^{j2\pi\frac{\alpha+\gamma-u}{M}x}}_{=M\delta_m(\alpha+\gamma-u)}\underbrace{\sum_{y} e^{j2\pi\frac{\beta+\delta-v}{N}y}}_{=N\delta_N(\beta+\delta-v)}$$

在求和中只有固定以下两个条件的项被保留下来：

$\gamma = u - \alpha \pmod M$　　$\delta = v - \beta \pmod N$

由于 γ 和 δ 取定等于上式的结果所以可以消掉两个和号：

$p(u,v) = \frac{1}{MN} \sum_{\alpha} \sum_{\beta} F(\alpha,\beta) H(u-\alpha, v-\beta)$

即 $p(u,v) = \frac{1}{MN} (F(u,v) * H(u,v))$

4.48* A continuous Gaussian lowpass filter in the continuous frequency domain has the transfer function

$$H(\mu,\nu) = Ae^{-(\mu^2+\nu^2)/2\sigma^2}$$

Show that the corresponding filter kernel in the continuous spatial domain is

$$h(t,z) = A2\pi\sigma^2 e^{-2\pi^2\sigma^2(t^2+z^2)}$$

CTFT:

$F(u,v) = \mathcal{F}[f(x,y)] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)\, e^{-j2\pi(ux+vy)}\, dx\,dy$

$f(x,y) = \mathcal{F}^{-1}[F(u,v)] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} F(u,v)\, e^{j2\pi(ux+vy)}\, du\,dv$

$h(t,z) = \mathcal{F}^{-1}[H(u,v)] = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} H(\mu,\nu)\, e^{j2\pi(\mu t+\nu z)}\, d\mu\,d\nu$

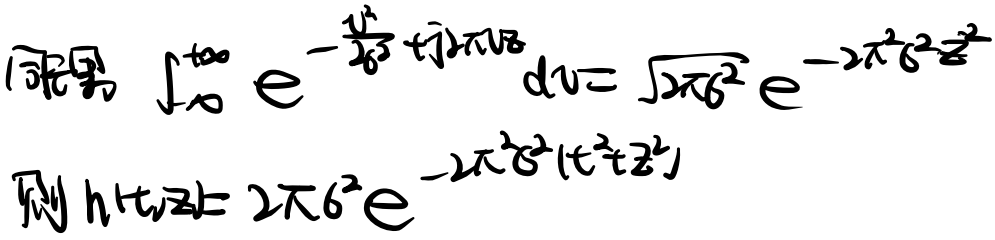
$= \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} A e^{-\frac{\mu^2+\nu^2}{2\sigma^2}} e^{j2\pi(\mu t+\nu z)}\, d\mu\,d\nu$

$= A\left(\int_{-\infty}^{+\infty} e^{-\frac{\mu^2}{2\sigma^2}+j2\pi\mu t}\, d\mu\right)\left(\int_{-\infty}^{+\infty} e^{-\frac{\nu^2}{2\sigma^2}+j2\pi\nu z}\, d\nu\right)$

$\int_{-\infty}^{+\infty} e^{-ax^2+bx}\, dx = \sqrt{\frac{\pi}{a}}\, e^{\frac{b^2}{4a}}$

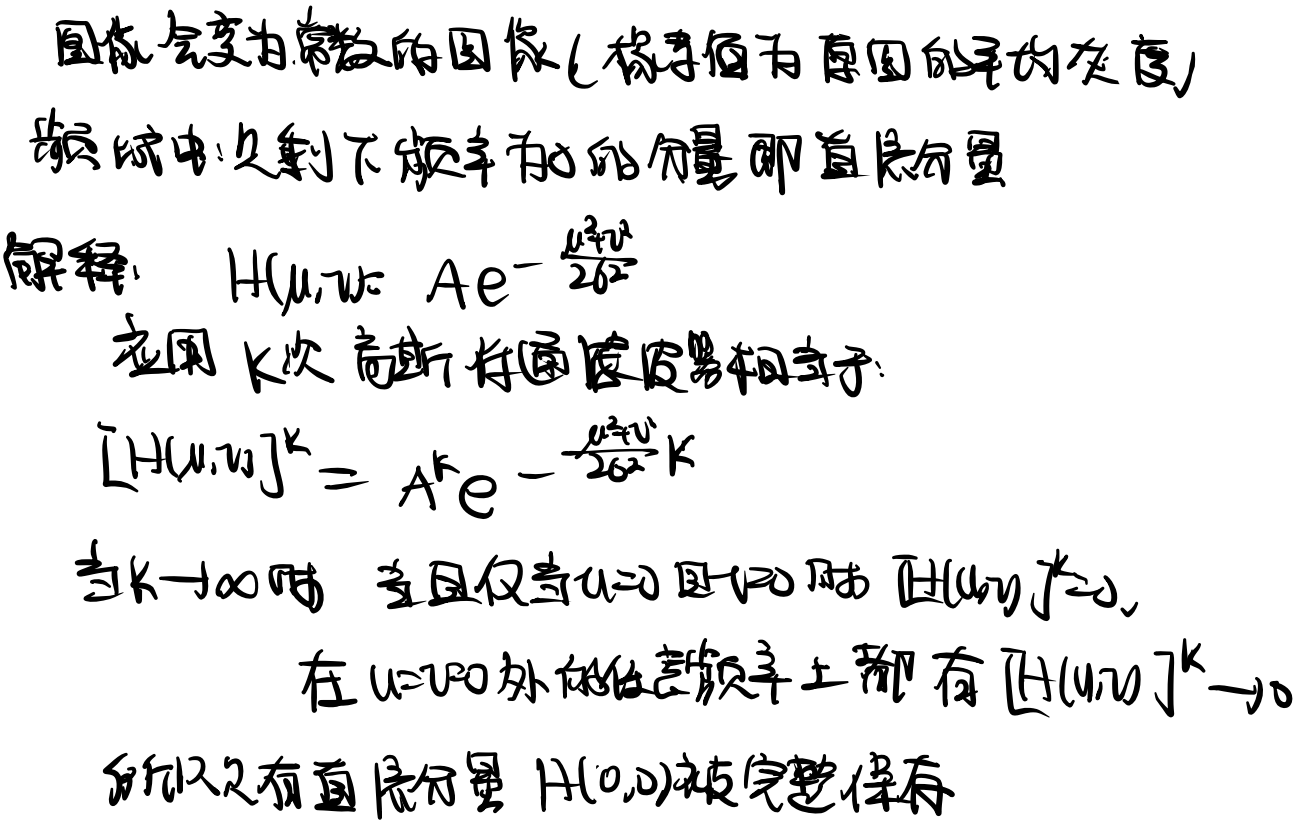
$\frac{-4\pi^2t^2\cdot 2\sigma^2}{4} = -2\pi^2\sigma^2t^2$

$\int_{-\infty}^{+\infty} e^{-\frac{\mu^2}{2\sigma^2}+j2\pi\mu t}\, d\mu = \sqrt{\frac{\pi}{\frac{1}{2\sigma^2}}}\, e^{\frac{(j2\pi t)^2}{\frac{4}{2\sigma^2}}} = \sqrt{2\pi\sigma^2}\, e^{-2\pi^2\sigma^2t^2}$



4.49 Given an image of size $M \times N$, you are asked to perform an experiment that consists of repeatedly lowpass filtering the image in the frequency domain using a Gaussian lowpass filter transfer function with a cutoff frequency, D_0. You may ignore computational round-off errors.

(a)* Let K denote the number of applications of the filter. Can you predict (without doing the experiment) what the result (image) will be for a sufficiently large value of K? If so, what is that result?



4.56* Show how the Butterworth highpass filter transfer function in Eq. (4-121) follows from its lowpass counterpart in Eq. (4-117).



$$令\left[\frac{D(u,v)}{D_0}\right]^{2n}=x$$

$$H_{hp}(u,v)=1-H_{lp}(u,v)=1-\frac{1}{1+\left[\frac{D(u,v)}{D_0}\right]^{2n}}=1-\frac{1}{1+x}$$

$$=\frac{1+x-1}{1+x}=\frac{x}{1+x}=\frac{1}{1+\frac{1}{x}}=\frac{1}{1+\left[\frac{D_0}{D(u,v)}\right]^{2n}}$$